

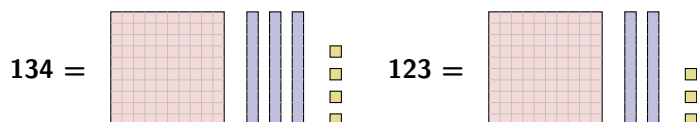
Introduction to SG Math

Base-Ten Blocks, Bar Models & Mental Math Strategies

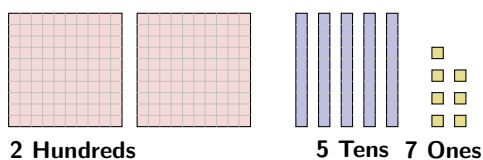
Part 1: Base-10 Block Addition

Example: Solve $134 + 123$ using Hundreds (Flats), Tens (Rods), and Ones (Units).

① Represent each number using blocks



② Group hundreds, tens, and ones together

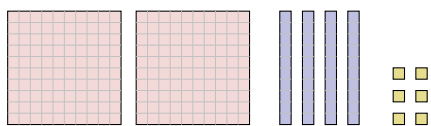


③ Write the final value

$$2 \text{ Hundreds} + 5 \text{ Tens} + 7 \text{ Ones} = 200 + 50 + 7 = \mathbf{257}$$

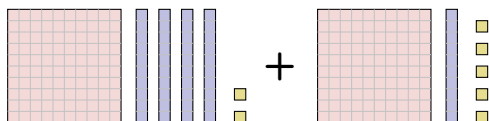
Practice Questions

1. What total number is shown by the blocks below?



_____ Hundreds + _____ Tens + _____ Ones = _____

2. Add the blocks to find $142 + 115$:



Total = _____ Hundreds + _____ Tens + _____ Ones = _____

Part 2: Base-10 Subtraction (Taking Away)

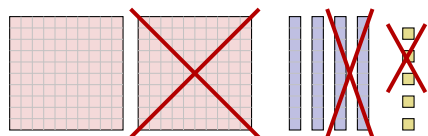
Example: Solve $245 - 123$ by crossing out blocks.

① **Build the starting number (245)**

Start with 2 Hundreds, 4 Tens, and 5 Ones.

② **Take away 123 (1 Hundred, 2 Tens, 3 Ones)**

Cross out the subtracted amounts with red lines:



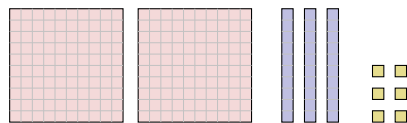
③ **Count remaining blocks**

Remaining = 1 Hundred, 2 Tens, 2 Ones

$$245 - 123 = 122$$

Practice Questions

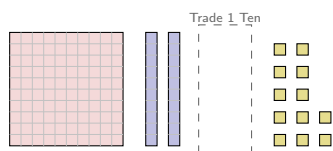
1. **Cross out blocks to solve** $236 - 114$:



Remaining: _____ Hundreds + _____ Tens + _____ Ones = _____

2. **Regrouping Tens to Ones:** $132 - 115$

Hint: Trade 1 Ten Rod for 10 Unit Cubes first, then take away 5 ones.



Equation: $132 - 115 =$ _____

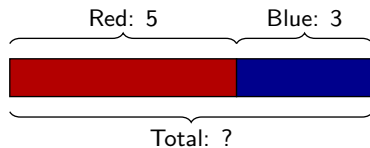
Part 3: Bar Models

Example: Maya has **5 red pencils** and **3 blue pencils**. How many pencils in total?

① **Identify the parts**

- Red pencils = 5
- Blue pencils = 3

② **Draw the bars**

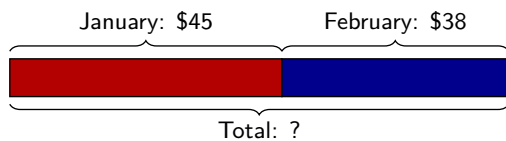


③ **Write and solve**

$$5 + 3 = 8$$

Practice Questions

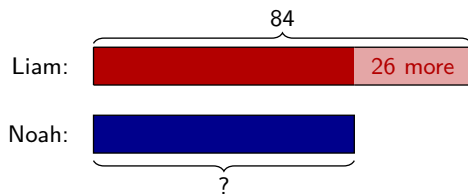
1. Lucas saved \$45 in January and \$38 in February. How much in total?



Equation: _____ + _____ = _____

Total = \$ _____

2. Liam has 84 stamps, 26 more than Noah. How many does Noah have?



Equation: _____ - _____ = _____

3. Apples : oranges is 4 : 7. There are 28 apples. How many oranges?

Apples: 

Oranges: 

1 unit = $28 \div 4 =$ _____

Oranges = $7 \times$ _____ = _____

Part 4: Mental Math Strategies (Friendly Numbers, Compensation & Shifting)

Example 1 (Compatible Numbers): Solve $128 + 245 + 273 + 155 + 27 + 72$ by grouping friendly pairs.

① **Identify compatible pairs**

Look at the ones and tens digits to pair numbers that make whole hundreds:

$$\boxed{128} + \boxed{245} + \boxed{273} + \boxed{155} + \boxed{27} + \boxed{72}$$

② **Rearrange and group the pairs**

$$(128 + 72) + (245 + 155) + (273 + 27)$$

③ **Add the friendly round hundreds**

$$200 + 400 + 300 = \mathbf{900}$$

Example 2 (Rounding & Compensation): Solve $999 + 99 + 9$ using friendly benchmark numbers.

① **Round up to the nearest powers of 10**

Each number is just 1 less than a round benchmark (1000, 100, 10):

$$\boxed{999} \xrightarrow{+1} \boxed{1000} \quad \boxed{99} \xrightarrow{+1} \boxed{100} \quad \boxed{9} \xrightarrow{+1} \boxed{10}$$

② **Add the round numbers and track the total excess**

We added $1 + 1 + 1 = 3$ extra, so we subtract 3 from the round total:

$$(1000 + 100 + 10) - 3 = 1110 - 3$$

③ **Calculate the final sum**

$$1110 - 3 = \mathbf{1107}$$

Example 3 (The Shifting Method for Subtraction): Solve $900 - 396$ by shifting to a round number.

① **Identify the nearest round hundred**

396 is only 4 away from 400.

② **Add the same amount (+4) to both numbers**

Adding the same number to both terms preserves the difference (no borrowing across zeroes!):

$$\boxed{900} - \boxed{396} \longrightarrow \boxed{904} - \boxed{400}$$

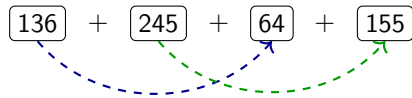
$\xrightarrow{+4}$ (from 900 to 904) and $\xrightarrow{+4}$ (from 396 to 400)

③ **Subtract without regrouping**

$$900 - 396 = 904 - 400 = \mathbf{504}$$

Practice Questions

1. Pair compatible numbers to make hundreds, then solve:



Equation: $(\underline{\hspace{2cm}} + \underline{\hspace{2cm}}) + (\underline{\hspace{2cm}} + \underline{\hspace{2cm}}) = \underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

2. Rearrange into three compatible pairs to solve $118 + 265 + 234 + 135 + 66 + 82$:

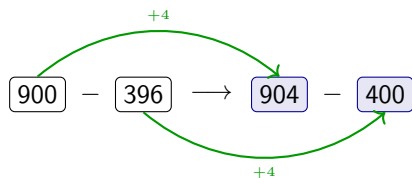
Equation: $(\underline{\hspace{2cm}} + \underline{\hspace{2cm}}) + (\underline{\hspace{2cm}} + \underline{\hspace{2cm}}) + (\underline{\hspace{2cm}} + \underline{\hspace{2cm}}) = \underline{\hspace{2cm}} + \underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

3. Use rounding and compensation to calculate $499 + 199 + 19$:

$$\boxed{499} \xrightarrow{+1} 500 \quad \boxed{199} \xrightarrow{+1} 200 \quad \boxed{19} \xrightarrow{+1} 20$$

Equation: $(\underline{\hspace{2cm}} + \underline{\hspace{2cm}} + \underline{\hspace{2cm}}) - \underline{\hspace{2cm}} = \underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

4. Use the shifting method (constant difference) to make a round subtrahend:



(a) $900 - 396 = (\underline{\hspace{2cm}} + 4) - (396 + 4) = 904 - 400 = \underline{\hspace{2cm}}$

(b) $800 - 297 = (800 + \underline{\hspace{2cm}}) - (297 + \underline{\hspace{2cm}}) = \underline{\hspace{2cm}} - 300 = \underline{\hspace{2cm}}$

(c) $600 - 189 = (600 + \underline{\hspace{2cm}}) - (189 + \underline{\hspace{2cm}}) = \underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

(d) $1000 - 495 = (1000 + \underline{\hspace{2cm}}) - (495 + \underline{\hspace{2cm}}) = \underline{\hspace{2cm}} - 500 = \underline{\hspace{2cm}}$